

RESEARCH ARTICLE

WILEY

Octonion short-time linear canonical transform

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Communicated by: T. Qian

Funding informationNational Key Research and Development
Project of China, Grant/Award Number:
2020YFC2006205

We investigate the octonion short-time linear canonical transform (OCSTLCT) in this paper. First, we propose the new definition of the OCSTLCT, and then several important properties of newly defined OCSTLCT, such as bounded, shift, modulation, time-frequency shift, inversion formula, and orthogonality relation, are derived based on the spectral representation of the octonion linear canonical transform (OCLCT). Second, by the Heisenberg uncertainty principle for the OCLCT and the orthogonality relation property for the OCSTLCT, the Heisenberg uncertainty principle for the OCSTLCT is established. Finally, we give an example of the OCSTLCT.

KEYWORDS

octonion, octonion linear canonical transform, octonion short-time linear canonical transform, uncertainty principle

MSC CLASSIFICATION

42A38, 11R52, 94A12

1 | INTRODUCTION

The linear canonical transform (LCT) has been found broad applications in many fields of applied mathematics, signal processing, radar system analysis, filter design, phase retrieval, and pattern recognition and optics.^{1–3} All of these results show that the LCT plays an important role for chirp signal analysis in parameter estimation, sampling, and filtering.^{4–8} However, the LCT can't show the local LCT-frequency contents as a result of its global kernel. The short-time linear canonical transform (STLCT) is a method devised to study signals whose spectral content changes in time. Some important properties of the STLCT have been discussed such as the analogue of the Poisson summation formula, sampling formulas, Paley–Wiener theorem, and uncertainty principles.^{9–13} It presents the time and LCT-frequency information, and is originally a local LCT distribution.

Quaternions (such as the Cayley–Dickson algebra of order 4) have wide applications in signal processing.^{14–17} Some authors have generalized the linear canonical transform to quaternion-valued signals, known as the quaternionic linear canonical transform (QLCT). The QLCT was firstly studied including prolate spheroidal wave signals and uncertainty principles by Bahri et al.^{18,19} Some useful properties of the QLCT such as linearity, reconstruction formula, continuity, boundedness, positivity inversion formula, and the uncertainty principle were established.^{20–23} An application of the QLCT to study of generalized swept-frequency filters was introduced.²³ Xiong and Fu²⁴ have been successfully applied the two-side QLCT to research the real windowed function, called the quaternion short-time linear canonical transform (QSTLCT), and several properties and Heisenberg's uncertainty principle for the QSTLCT have been studied. Different from Xiong and Fu,²⁴ based on the quaternion window function, we²⁵ have investigated some properties of the two-side QSTLCT, and gave an example to illustrative the QSTLCT is highly effective. Recently, based on some properties of the

two-side QSTLCT, we²⁶ obtain different kinds of uncertainty principles for the QSTLCT. Moreover, Teali²⁷ has been proposed the short-time offset linear canonical transform for quaternion-valued signals and studied some properties.

On the other hand, octonions (such as the Cayley–Dickson algebra of order 8) are also known as Cayley numbers and are another hypercomplex algebra. Based on the definition of octonion algebra and its algebraic properties, Kauhanen²⁸ derived e_4 -calculus and obtained a list of generalized Cauchy Riemann systems for octonionic monogenic functions. Hahn and Snopek²⁹ presented selected algebras of basis vectors for 3-D octonionic signals, and the hypothesis regarding the polar form of an octonionic signal was verified based on numerical examples. However, Błaszczuk³⁰ disproved the above result and proposed a novel numerical method. The presented numerical method gives promising results, which as a result of further work can give analytical formulas for angles in the polar representation.³⁰ Snopek³¹ proposed the octonion Fourier transform (OFT) of the real-valued function of three variables. The OFT is constructed by replacing the quaternion kernel function with the octonion kernel function. As a generalization of the quaternion Fourier transform (QFT), researchers have been interested in the OFT for many years. The properties of the OFT of real-valued functions of three variables and its potential applications in signal and system processing have been investigated.^{32,33} Snopek³⁴ derived symmetry relations, shift properties, and the Parseval and Plancherel theorems for the OFT of real-valued functions. Lian³⁵ exploited several uncertainty inequalities of the OFT. The properties of the OFT in the context of signal-domain operations such as derivation and convolution were also stated.³⁶ Recently, Błaszczuk³⁷ proposed the discrete OFT, which is an excellent tool for the analysis of 3-D discrete linear time invariant (LTI) systems and 3-D discrete data. We³⁸ introduced the octonion linear canonical transform (OCLCT), which is generalization of the OFT in the framework of octonion algebra. Several properties for the OCLCT such as shift property, inversion formula, isometry, Riemann–Lebesgue lemma, and uncertainty principles are explored.³⁸

The OCLC converts a 3-D signal from the space domain to the corresponding spectral domain. But it cannot reveal the local frequency contents due to its global kernel. However, it is believed that carry out the analysis of the 3-D signal in both the space and frequency domains simultaneously is very useful and important for applications of signal processing. To tackle this problem, by adding a suitable window to the OCLCT, we propose a novel transform tool-octonion short-time linear canonical transform (OCSTLCT) in this paper, which is also called octonion window linear canonical transform and can be used to efficiently extract information from three-dimensional data. The main goals of the present paper are to present the novel concept of the OCSTLCT and research important properties of the OCSTLCT such as boundedness, linearity, parity, shift, modulation, time-frequency shifts, inversion formula, and orthogonality relation. Additionally, according to the Heisenberg uncertainty principle for the OCLCT and the orthogonality relation property for the OCSTLCT, we establish a generalized uncertainty principle associated with OCSTLCT. This also lays a good foundation for the practical application in the future. Finally, we present an example of the OCSTLCT.

The paper is organized as follows: Section 2 gives a brief introduction to some general definitions and basic properties of octonion algebra. We give the definition and study the properties of the OCSTLCT in Section 3. Section 4 provides the uncertainty principle associated with the OCSTLCT. We give an example of the OCSTLCT in Section 5. In Section 6, some conclusions are drawn.

2 | PRELIMINARIES

In this section, we mainly review some basic facts regarding octonion algebra and the definitions of the OCLCT, which will be needed throughout the paper.

2.1 | Octonion algebra

According to the Cayley–Dickson construction,³⁷ an arbitrary $o \in \mathbb{O}$ can be written in the form

$$o = o_0 + o_1e_1 + o_2e_2 + o_3e_3 + o_4e_4 + o_5e_5 + o_6e_6 + o_7e_7,$$

where $o_0, o_1, \dots, o_7 \in \mathbb{R}$ and $e_1, e_2, \dots, e_7$ are seven different imaginary units. o_0 is called the real part of o , and the pure imaginary octonion $o_1e_1 + o_2e_2 + o_3e_3 + o_4e_4 + o_5e_5 + o_6e_6 + o_7e_7$ is called the imaginary part of o . The octonion algebra is a non-associative and non-commutative algebra. The multiplication rules in octonion algebra are presented in Table 1.

The octonion conjugate is defined by

$$\bar{o} = o_0 - o_1e_1 - o_2e_2 - o_3e_3 - o_4e_4 - o_5e_5 - o_6e_6 - o_7e_7.$$

*	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
1	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	-1	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	-1	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	-1	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	-1	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	-1	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	-1	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	-1

TABLE 1 Multiplication rules in octonion algebra

We can define the norm by $|o| = \sqrt{o\bar{o}} = \sqrt{\bar{o}o}$ and $|o|^2 = \sum_{j=0}^7 o_j^2$. The norm satisfies $|\epsilon\epsilon| = |\epsilon||\epsilon|$, for all $\epsilon, \epsilon \in \mathbb{O}$. The exponential function of an octonion variable is defined by

$$e^o = \sum_{k=0}^{\infty} \frac{o^k}{k!}.$$

Every octonion $o \in \mathbb{O}$ can be represented in the form

$$o = \alpha + \beta e_4,$$

where $\alpha = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 \in \mathbb{H}$ and $\beta = o_4 + o_5 e_1 + o_6 e_2 + o_7 e_3 \in \mathbb{H}$. This form is called the quaternionic form of an octonion.

Moreover, we have the following properties.^{28,34}

Property 1. Let $\alpha, \beta \in \mathbb{H}$. Then

$$(1)e_4\alpha = \bar{\alpha}e_4; (2)e_4(\alpha e_4) = -\bar{\alpha}; (3)(\alpha e_4)e_4 = -\alpha; (4)\alpha(\beta e_4) = (\beta\alpha)e_4; (5)(\alpha e_4)\beta = (\alpha\bar{\beta})e_4.$$

Property 2. For an octonion $\alpha + \beta e_4$, $\alpha, \beta \in \mathbb{H}$ in the quaternionic form we have

$$\overline{\alpha + \beta e_4} = \bar{\alpha} - \beta e_4, |\alpha + \beta e_4|^2 = |\alpha|^2 + |\beta|^2.$$

Property 3. Let $\bar{o}, \hat{o} \in \mathbb{O}$. Then

$$e^{\bar{o}} \cdot e^{\hat{o}} = e^{\bar{o}+\hat{o}} \text{ if and only if } \bar{o} \cdot \hat{o} = \hat{o} \cdot \bar{o}.$$

For each real-valued function $f(x)$ over $\mathbb{R}^d$ and $1 \leq p < \infty$, the L^p -norm of f is defined by

$$\|f\|_{L^p(\mathbb{R}^d)} = \left(\int_{\mathbb{R}^d} |f(\mathbf{x})|^p d\mathbf{x} \right)^{\frac{1}{p}},$$

where $\mathbf{x} = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, if $p = \infty$, then the L^∞ -norm of f is defined by

$$\|f\|_\infty = \text{ess sup}_{\mathbf{x} \in \mathbb{R}^d} |f(\mathbf{x})|.$$

An inner product of real-valued functions $f(\mathbf{x})$, $g(\mathbf{x})$ is given by

$$\langle f, g \rangle = \int_{\mathbb{R}^d} f(\mathbf{x}) \overline{g(\mathbf{x})} d\mathbf{x}.$$

Let $f, g \in L^2(\mathbb{R}^d)$, the classic convolution operation is defined as

$$(f * g)(\mathbf{x}) = \int_{\mathbb{R}^d} f(\mathbf{y})g(\mathbf{x} - \mathbf{y})d\mathbf{y}.$$

2.2 | Octonion linear canonical transform

In the following part, we introduce the OCLCT.³⁸ Unless otherwise stated, we assume that the following signal is continuous and that both the signal and its OCLCT are integrable (in the Lebesgue sense) in this paper. Moreover, it is necessary to note that the multiplication in the next integral is done from left to right because the octonion algebra is nonassociative.

Definition 1. Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2, 3$. The OCLCT of signal $f \in L^2(\mathbb{R}^3)$ is defined by

$$\mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{w}) = \int_{\mathbb{R}^3} f(\mathbf{x}) K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x}, \quad (1)$$

where $\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$, $\mathbf{w} = (\omega_1, \omega_2, \omega_3) \in \mathbb{R}^3$ and the kernel signals

$$K_{A_1}^{e_1}(x_1, \omega_1) := \begin{cases} \frac{1}{\sqrt{2\pi|b_1|}} e^{e_1 \left(\frac{a_1}{2b_1} x_1^2 - \frac{x_1 \omega_1}{b_1} + \frac{d_1}{2b_1} \omega_1^2 - \frac{\pi}{2} \right)}, & b_1 \neq 0 \\ \sqrt{d_1} e^{e_1 \frac{c_1 d_1}{2} \omega_1^2} \delta(x_1 - d_1 \omega_1), & b_1 = 0 \end{cases}, \quad (2)$$

$$K_{A_2}^{e_2}(x_2, \omega_2) := \begin{cases} \frac{1}{\sqrt{2\pi|b_2|}} e^{e_2 \left(\frac{a_2}{2b_2} x_2^2 - \frac{x_2 \omega_2}{b_2} + \frac{d_2}{2b_2} \omega_2^2 - \frac{\pi}{2} \right)}, & b_2 \neq 0 \\ \sqrt{d_2} e^{e_2 \frac{c_2 d_2}{2} \omega_2^2} \delta(x_2 - d_2 \omega_2), & b_2 = 0 \end{cases}, \quad (3)$$

and

$$K_{A_3}^{e_4}(x_3, \omega_3) := \begin{cases} \frac{1}{\sqrt{2\pi|b_3|}} e^{e_4 \left(\frac{a_3}{2b_3} x_3^2 - \frac{x_3 \omega_3}{b_3} + \frac{d_3}{2b_3} \omega_3^2 - \frac{\pi}{2} \right)}, & b_3 \neq 0 \\ \sqrt{d_3} e^{e_4 \frac{c_3 d_3}{2} \omega_3^2} \delta(x_3 - d_3 \omega_3), & b_3 = 0 \end{cases}, \quad (4)$$

where $\delta(\mathbf{x})$ representing the Dirac function.

From the above definition, it is noted that for $b_i = 0, i = 1, 2, 3$ the OCLCT of a signal is a kind of scaling and chirp multiplication operations, and it is of no particular interest for our objective in this work. Hence, without loss of generality, we set $b_i \neq 0$ in the following section unless stated otherwise. Under some suitable conditions, the OCLCT above is invertible and the inversion is given in the following section.

Lemma 1. [Inversion formula of the OCLCT] Let a real valued function $f \in L^2(\mathbb{R}^3)$. Then we have the inversion formula of the OCLCT,

$$\begin{aligned} f(\mathbf{x}) &= (\mathcal{L}_{A_1, A_2, A_3}^o)^{-1} [\mathcal{L}_{A_1, A_2, A_3}^o \{f\}](\mathbf{x}) \\ &= \int_{\mathbb{R}^3} \mathcal{L}_{A_1, A_2, A_3}^o \{f\}(\mathbf{w}) K_{A_3^{-1}}^{e_4}(\omega_3, x_3) K_{A_2^{-1}}^{e_2}(\omega_2, x_2) K_{A_1^{-1}}^{e_1}(\omega_1, x_1) d\mathbf{w}, \end{aligned} \quad (5)$$

where $A_i^{-1} = \begin{bmatrix} d_i & -b_i \\ -c_i & a_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, for $i = 1, 2, 3$.

The Parseval formula of OCLCT is very available as follows:

Lemma 2 [Parseval theorem of the OCLCT]. Let a real valued function $f \in L^2(\mathbb{R}^3)$ be a continuous and square integrable function. Then we have

$$\|f\|_{L^2(\mathbb{R}^3)}^2 = 2\pi|b_3| \|\mathcal{L}_{A_1, A_2, A_3}^o \{f\}\|_{L^2(\mathbb{R}^3)}^2. \quad (6)$$

3 | OCTONION SHORT-TIME LINEAR CANONICAL TRANSFORM

In this section, the generalization of window function associated with the OCLCT will be discussed, which is denoted by octonion short-time linear canonical transform (OCSTLCT). Moreover, several basic properties of them are investigated. Let e_1, e_2 and e_3 (or equivalently i, j, k) denote the three imaginary units in the quaternion algebra.³⁴

Definition 2. Let a real-valued function $f \in L^2(\mathbb{R}^3)$ and a real-valued window function $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$. Then the OCSTLCT of f is defined as

$$G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) = \int_{\mathbb{R}^3} f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x}, \quad (7)$$

where $\mathbf{x} = (x_1, x_2, x_3)$, $\mathbf{w} = (w_1, w_2, w_3)$ and $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$. $\mathbf{w}$ and $\mathbf{u}$ denote frequency and time, respectively. $\bar{\phi}$ denotes the complex conjugate of ϕ because the window function ϕ is real-valued, so $\bar{\phi} = \phi$.

For a fixed $\mathbf{u}$, we have

$$G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) = \mathcal{L}_{A_1, A_2, A_3}^o \{f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})}\}(\mathbf{w}). \quad (8)$$

Applying Lemma 1 to (8), we have

$$f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} = \int_{\mathbb{R}^3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) K_{A_3}^{e_4}(\omega_3, x_3) K_{A_2}^{e_2}(\omega_2, x_2) K_{A_1}^{e_1}(\omega_1, x_1) d\mathbf{w}. \quad (9)$$

According to the previous work,³⁸ the kernel of the OCSTLCT can be expanded in the form

$$\begin{aligned} & \left(K_{A_1}^{e_1}(x_1, w_1) K_{A_2}^{e_2}(x_2, w_2) \right) K_{A_3}^{e_4}(x_3, w_3) \\ &= \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} (\cos \gamma_1 \cos \gamma_2 \cos \gamma_3 + \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 e_1 \\ & \quad + \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 e_2 + \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 e_3 + \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 e_4 \\ & \quad + \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 e_5 + \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 e_6 + \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 e_7), \end{aligned} \quad (10)$$

where $\gamma_k = \frac{a_k}{2b_k} x_k^2 - \frac{x_k w_k}{b_k} + \frac{d_k}{2b_k} w_k^2 - \frac{\pi}{2}$, $k = 1, 2, 3$. The form (10) is a full octonion as defined in (7). It is worth noticing that in the definition of the OCSTLCT one can use any of the 28 octonion triples that doesn't form an associative, quaternion, non-commutative subfield of $\mathbb{O}$.

Based on (10), the OCSTLCT of a real valued function f of three variables can be expressed, as follows:

$$\begin{aligned} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) &= G_{eee}^{A_1, A_2, A_3} + G_{oe e}^{A_1, A_2, A_3} e_1 + G_{eoe}^{A_1, A_2, A_3} e_2 + G_{ooo}^{A_1, A_2, A_3} e_3 + G_{eeo}^{A_1, A_2, A_3} e_4 \\ & \quad + G_{oeo}^{A_1, A_2, A_3} e_5 + G_{eoo}^{A_1, A_2, A_3} e_6 + G_{ooo}^{A_1, A_2, A_3} e_7, \end{aligned} \quad (11)$$

where

$$\begin{aligned}
 G_{eee}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eee}(\mathbf{x}) \overline{\phi_{eee}(\mathbf{x} - \mathbf{u})} \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x}; \\
 G_{oee}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{oee}(\mathbf{x}) \overline{\phi_{oee}(\mathbf{x} - \mathbf{u})} \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x}; \\
 G_{eoe}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eoe}(\mathbf{x}) \overline{\phi_{eoe}(\mathbf{x} - \mathbf{u})} \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 d\mathbf{x}; \\
 G_{ooo}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{ooo}(\mathbf{x}) \overline{\phi_{ooo}(\mathbf{x} - \mathbf{u})} \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 d\mathbf{x}; \\
 G_{eeo}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eeo}(\mathbf{x}) \overline{\phi_{eeo}(\mathbf{x} - \mathbf{u})} \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 d\mathbf{x}; \\
 G_{oeo}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{oeo}(\mathbf{x}) \overline{\phi_{oeo}(\mathbf{x} - \mathbf{u})} \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 d\mathbf{x}; \\
 G_{eoo}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eoo}(\mathbf{x}) \overline{\phi_{eoo}(\mathbf{x} - \mathbf{u})} \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 d\mathbf{x}; \\
 G_{ooo}^{A_1, A_2, A_3}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{ooo}(\mathbf{x}) \overline{\phi_{ooo}(\mathbf{x} - \mathbf{u})} \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 d\mathbf{x},
 \end{aligned}$$

where $B = \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}}$.

We use subscripts e and o to indicate that a function is either even (e) or odd (o) with respect to an appropriate variable, that is, $f_{oe}(\mathbf{x})$ is odd with respect to x_1 and x_2 and even with respect to x_3 .

3.1 | Some properties of OCSTLCT

In this subsection, we discuss several basic properties of the OCSTLCT. These properties play important roles in signal representation.

Theorem 1 [Boundedness]. Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a window function and $f \in L^2(\mathbb{R}^3)$, then

$$|G_{o,\phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u})| \leq \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \|f\|_{L^2(\mathbb{R}^3)} \|\phi\|_{L^2(\mathbb{R}^3)}. \quad (12)$$

Proof. By using the quaternion Cauchy–Schwarz inequality

$$\begin{aligned}
 |G_{o,\phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u})|^2 &= \left| \int_{\mathbb{R}^3} f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x} \right|^2 \\
 &\leq \left(\int_{\mathbb{R}^3} |f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3)| d\mathbf{x} \right)^2 \\
 &= \left(\frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} |f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})}| d\mathbf{x} \right)^2 \\
 &\leq \frac{1}{8\pi^3|b_1b_2b_3|} \left(\int_{\mathbb{R}^3} |f(\mathbf{x})|^2 d\mathbf{x} \right) \left(\int_{\mathbb{R}^3} |\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} \right) \\
 &= \frac{1}{8\pi^3|b_1b_2b_3|} \|f\|_{L^2(\mathbb{R}^3)}^2 \|\phi\|_{L^2(\mathbb{R}^3)}^2,
 \end{aligned} \quad (13)$$

where applying the change of variables $\mathbf{x} - \mathbf{u} = \mathbf{t}$ in the last step. Then we have

$$|G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u})| \leq \frac{1}{2\pi \sqrt{2\pi |b_1 b_2 b_3|}} \|f\|_{L^2(\mathbb{R}^3)} \|\phi\|_{L^2(\mathbb{R}^3)}, \quad (14)$$

which completes the proof. $\square$

Theorem 2 [Linearity]. Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a window function and $f, g \in L^2(\mathbb{R}^3)$. The OCSTLCT is a linear operator, namely,

$$[G_{o,\phi}^{A_1,A_2,A_3} (\lambda f + \mu g)](\mathbf{w}, \mathbf{u}) = \lambda G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) + \mu G_{o,\phi}^{A_1,A_2,A_3} g(\mathbf{w}, \mathbf{u}), \quad (15)$$

for arbitrary real constants λ and μ .

Proof. By the product and the integration involved in Definition 2, we can obtain the result. $\square$

Theorem 3 [Parity]. Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a window function and $f \in L^2(\mathbb{R}^3)$. Then we have

$$G_{o,P\phi}^{A_1,A_2,A_3} \{Pf\}(\mathbf{w}, \mathbf{u}) = G_{o,\phi}^{A_1,A_2,A_3} f(-\mathbf{w}, -\mathbf{u}) \quad (16)$$

where $P\phi(\mathbf{x}) = \phi(-\mathbf{x})$ for every window function $\phi \in L^2(\mathbb{R}^2)$.

Proof. According to Definition 2, we obtain

$$G_{o,P\phi}^{A_1,A_2,A_3} \{Pf\}(\mathbf{w}, \mathbf{u}) = \int_{\mathbb{R}^3} f(-\mathbf{x}) \overline{\phi(-(\mathbf{x} - \mathbf{u}))} K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) dx. \quad (17)$$

Based on the fact

$$\begin{aligned} K_{A_1}^{e_1}(x_1, \omega_1) &= K_{A_1}^{e_1}(-x_1, -\omega_1); \\ K_{A_2}^{e_2}(x_2, \omega_2) &= K_{A_2}^{e_2}(-x_2, -\omega_2); \\ K_{A_3}^{e_4}(x_3, \omega_3) &= K_{A_3}^{e_4}(-x_3, -\omega_3), \end{aligned} \quad (18)$$

we can obtain the result. $\square$

Let $T_{r_1} f(\mathbf{x}) = f(x_1 - r_1, x_2, x_3)$, it shows that the function shifted with respect to the first variable. The shift with respect to the first variable does not change the parity with respect to other remaining variables. The shift theorem of the OCSTLCT can be obtained as follows.

Theorem 4 [Shift]. Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a window function and $f \in L^2(\mathbb{R}^3)$. Assume that $r_k, r'_k \in \mathbb{R}$, ($k = 1, 2, 3$) and $T_{r_1} f(\mathbf{x}) = f(x_1 - r_1, x_2, x_3)$, $T_{r'_1} \phi(\mathbf{x}) = \phi(x_1 - r'_1, x_2, x_3)$, $T_{r_2} f(\mathbf{x}) = f(x_1, x_2 - r_2, x_3)$, $T_{r'_2} \phi(\mathbf{x}) = \phi(x_1, x_2 - r'_2, x_3)$, $T_{r_3} f(\mathbf{x}) = f(x_1, x_2, x_3 - r_3)$ and $T_{r'_3} \phi(\mathbf{x}) = \phi(x_1, x_2, x_3 - r'_3)$. Then we have

$$G_{o,T_{r'_1}\phi}^{A_1,A_2,A_3} T_{r_1} f(\mathbf{w}, \mathbf{u}) = \cos \phi_1^{A_1,A_2,A_3} G_{o,\phi}^{A_1,A_2,A_3} f(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1,A_2,A_3} \Theta_1^{A_1,A_2,A_3}(\tilde{\mathbf{w}}, \mathbf{u}); \quad (19)$$

$$G_{o,T_{r'_2}\phi}^{A_1,A_2,A_3} T_{r_2} f(\mathbf{w}, \mathbf{u}) = \cos \phi_2^{A_1,A_2,A_3} G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) - \sin \phi_2^{A_1,A_2,A_3} \Theta_2^{A_1,A_2,A_3}(\mathbf{w}, \mathbf{u}); \quad (20)$$

$$G_{o,T_{r'_3}\phi}^{A_1,A_2,A_3} T_{r_3} f(\mathbf{w}, \mathbf{u}) = \cos \phi_3^{A_1,A_2,A_3} G_{o,\phi}^{A_1,A_2,A_3} f(\bar{\mathbf{w}}, \mathbf{u}) - \sin \phi_3^{A_1,A_2,A_3} \Theta_3^{A_1,A_2,A_3}(\bar{\mathbf{w}}, \mathbf{u}), \quad (21)$$

where $\tilde{\mathbf{w}} = (m_1, w_2, w_3)$, $\mathbf{w} = (w_1, m_2, w_3)$, $\bar{\mathbf{w}} = (w_1, w_2, m_3)$, $m_k = w_k - r_k a_k$, $\phi_k^{A_1,A_2,A_3} = r_k w_k c_k - \frac{a_k c_k r_k^2}{2}$ ($k = 1, 2, 3$) and $\Theta_1^{A_1,A_2,A_3} = G_{see}^{A_1,A_2,A_3} - G_{cee}^{A_1,A_2,A_3} e_1 + G_{soe}^{A_1,A_2,A_3} e_2 - G_{coe}^{A_1,A_2,A_3} e_3 + G_{seo}^{A_1,A_2,A_3} e_4 - G_{ceo}^{A_1,A_2,A_3} e_5 + G_{soo}^{A_1,A_2,A_3} e_6 - G_{coo}^{A_1,A_2,A_3} e_7$, $\Theta_2^{A_1,A_2,A_3} = G_{ese}^{A_1,A_2,A_3} + G_{ose}^{A_1,A_2,A_3} e_1 - G_{ece}^{A_1,A_2,A_3} e_2 - G_{oce}^{A_1,A_2,A_3} e_3 + G_{eso}^{A_1,A_2,A_3} e_4 + G_{oso}^{A_1,A_2,A_3} e_5 - G_{eco}^{A_1,A_2,A_3} e_6 - G_{oco}^{A_1,A_2,A_3} e_7$, $\Theta_3^{A_1,A_2,A_3} = G_{ees}^{A_1,A_2,A_3} + G_{oes}^{A_1,A_2,A_3} e_1 + G_{eos}^{A_1,A_2,A_3} e_2 + G_{oos}^{A_1,A_2,A_3} e_3 - G_{eec}^{A_1,A_2,A_3} e_4 - G_{oec}^{A_1,A_2,A_3} e_5 - G_{eoc}^{A_1,A_2,A_3} e_6 - G_{ooc}^{A_1,A_2,A_3} e_7$.

Proof. The OCSTLCT of the function $T_{r_1} f(\mathbf{x})$ is expressed by (11) as follows:

$$G_{o,T_{r_1} f}^{A_1, A_2, A_3} T_{r_1} f(\mathbf{w}, \mathbf{u}) = G_{eee}^{T_{r_1}'} + G_{oe e}^{T_{r_1}'} e_1 + G_{eoe}^{T_{r_1}'} e_2 + G_{ooe}^{T_{r_1}'} e_3 + G_{e eo}^{T_{r_1}'} e_4 + G_{oeo}^{T_{r_1}'} e_5 + G_{e oo}^{T_{r_1}'} e_6 + G_{ooo}^{T_{r_1}'} e_7. \quad (22)$$

Next we will calculate $G_{eee}^{T_{r_1}'}$. According to (11), then

$$\begin{aligned} G_{eee}^{T_{r_1}'}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} T_{r_1} f_{eee}(\mathbf{x}) \overline{T_{r_1}' \phi_{eee}(\mathbf{x} - \mathbf{u})} \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x} \\ &= B \int_{\mathbb{R}^3} f_{eee}(x_1 - r_1, x_2, x_3) \overline{\phi_{eee}(x_1 - u_1 - r_1', x_2 - u_2, x_3 - u_3)} \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 d\mathbf{x}, \end{aligned} \quad (23)$$

where $B = \frac{1}{2\pi\sqrt{2\pi|b_1 b_2 b_3|}}$. By making the change of variables $\mathbf{y} = (y_1, y_2, y_3) = (x_1 - r_1, x_2, x_3)$, $\mathbf{z}_{r_1} = (r_1' + u_1 - r_1, u_2, u_3)$ and $\mathbf{y} - \mathbf{z}_{r_1} = (y_1 - (r_1' + u_1 - r_1), y_2 - u_2, y_3 - u_3)$ in the above expression, we obtain

$$\begin{aligned} G_{eee}^{T_{r_1}'}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eee}(\mathbf{y}) \overline{\phi_{eee}(\mathbf{y} - \mathbf{z}_{r_1})} \cos(\phi_1^{A_1, A_2, A_3} + \varphi_1) \cos \psi_2 \cos \psi_3 d\mathbf{y} \\ &= B \left[\cos \phi_1^{A_1, A_2, A_3} \int_{\mathbb{R}^3} f_{eee}(\mathbf{y}) \overline{\phi_{eee}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y} \right. \\ &\quad \left. - \sin \phi_1^{A_1, A_2, A_3} \int_{\mathbb{R}^3} f_{eee}(\mathbf{y}) \overline{\phi_{eee}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y} \right], \end{aligned} \quad (24)$$

where $\varphi_1 = \frac{a_1}{2b_1} y_1^2 - \frac{y_1(w_1 - r_1 a_1)}{b_1} + \frac{d_1}{2b_1} (w_1 - r_1 a_1)^2 - \frac{\pi}{2}$, $\psi_k = \frac{a_k}{2b_k} y_k^2 - \frac{y_k w_k}{b_k} + \frac{d_k}{2b_k} w_k^2 - \frac{\pi}{2}$, $k = 2, 3$.

Let

$$\begin{aligned} G_{see}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eee}(\mathbf{y}) \overline{\phi_{eee}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y}, \\ G_{cee}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{oe e}(\mathbf{y}) \overline{\phi_{oe e}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y}, \\ G_{soe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{eoe}(\mathbf{y}) \overline{\phi_{eoe}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \sin \psi_2 \cos \psi_3 d\mathbf{y}, \\ G_{coe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{ooe}(\mathbf{y}) \overline{\phi_{ooe}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \sin \psi_2 \cos \psi_3 d\mathbf{y}, \\ G_{seo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{e eo}(\mathbf{y}) \overline{\phi_{e eo}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \cos \psi_2 \sin \psi_3 d\mathbf{y}, \\ G_{ceo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{oeo}(\mathbf{y}) \overline{\phi_{oeo}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \cos \psi_2 \sin \psi_3 d\mathbf{y}, \\ G_{soo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{e oo}(\mathbf{y}) \overline{\phi_{e oo}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \sin \psi_2 \sin \psi_3 d\mathbf{y}, \\ G_{coo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{ooo}(\mathbf{y}) \overline{\phi_{ooo}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \sin \psi_2 \sin \psi_3 d\mathbf{y}. \end{aligned}$$

Then, the formula (24) can be rewritten in the form

$$G_{eee}^{T_1^1}(\mathbf{w}, \mathbf{u}) = \cos \phi_1^{A_1, A_2, A_3} G_{eee}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} G_{see}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}). \quad (25)$$

Similarly, we can prove that

$$\begin{aligned} G_{oeo}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= B \int_{\mathbb{R}^3} f_{oeo}(\mathbf{y}) \overline{\phi_{oeo}(\mathbf{y} - \mathbf{z}_{r_1})} \sin(\phi_1^{A_1, A_2, A_3} + \varphi_1) \cos \psi_2 \cos \psi_3 d\mathbf{y} \\ &= B \left[\sin \phi_1^{A_1, A_2, A_3} \int_{\mathbb{R}^3} f_{oeo}(\mathbf{y}) \overline{\phi_{oeo}(\mathbf{y} - \mathbf{z}_{r_1})} \cos \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y} \right. \\ &\quad \left. + \cos \phi_1^{A_1, A_2, A_3} \int_{\mathbb{R}^3} f_{oeo}(\mathbf{y}) \overline{\phi_{oeo}(\mathbf{y} - \mathbf{z}_{r_1})} \sin \varphi_1 \cos \psi_2 \cos \psi_3 d\mathbf{y} \right] \\ &= \cos \phi_1^{A_1, A_2, A_3} G_{oeo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) + \sin \phi_1^{A_1, A_2, A_3} G_{coe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}). \end{aligned}$$

Summarizing, we have

$$\begin{aligned} G_{eee}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{eee}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} G_{see}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}); \\ G_{oeo}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{oeo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) + \sin \phi_1^{A_1, A_2, A_3} G_{coe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}); \\ G_{eoe}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{eoe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} G_{soe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}); \\ G_{ooe}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{ooe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) + \sin \phi_1^{A_1, A_2, A_3} G_{coe}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}), \\ G_{eoo}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{eoo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} G_{seo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}), \\ G_{oee}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{oee}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) + \sin \phi_1^{A_1, A_2, A_3} G_{ceo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}), \\ G_{ooo}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{ooo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} G_{soo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}), \\ G_{ooo}^{T_1^1}(\mathbf{w}, \mathbf{u}) &= \cos \phi_1^{A_1, A_2, A_3} G_{ooo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}) + \sin \phi_1^{A_1, A_2, A_3} G_{coo}^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}). \end{aligned}$$

Applying (23), we finally arrive at

$$G_{o, T_1^1 \phi}^{A_1, A_2, A_3} T_{r_1} f(\mathbf{w}, \mathbf{u}) = \cos \phi_1^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\tilde{\mathbf{w}}, \mathbf{u}) - \sin \phi_1^{A_1, A_2, A_3} \Theta_1^{A_1, A_2, A_3}(\tilde{\mathbf{w}}, \mathbf{u}).$$

Analogously method hold for concerning shifts with respect to variables x_2 and x_3 . $\square$

Theorem 5. Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that for all $w_0, w' \in \mathbb{R}$, $\zeta_i = 2\pi w_0 x_i$, $\zeta'_i = 2\pi w' x_i$ ($i = 1, 2, 3$), $M_{s,1}^{w_0} f(\mathbf{x}) = f(\mathbf{x}) \sin \zeta_1$, $M_{s,1}^{w'} \phi(\mathbf{x}) = \phi(\mathbf{x}) \sin \zeta'_1$, $M_{s,2}^{w_0} f(\mathbf{x}) = f(\mathbf{x}) \sin \zeta_2$, $M_{s,2}^{w'} \phi(\mathbf{x}) = \phi(\mathbf{x}) \sin \zeta'_2$, $M_{s,3}^{w_0} f(\mathbf{x}) = f(\mathbf{x}) \sin \zeta_3$ and $M_{s,3}^{w'} \phi(\mathbf{x}) = \phi(\mathbf{x}) \sin \zeta'_3$. Then

$$\begin{aligned} G_{o, M_{s,1}^{w_0} \phi}^{A_1, A_2, A_3} (M_{s,1}^{w_0} f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(\cos(\Phi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,1}^1, \mathbf{u}) + \cos(\Xi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,2}^1, \mathbf{u}) \right. \\ &\quad \left. - \cos(\Pi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^1, \mathbf{u}) - \cos(\Lambda_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^1, \mathbf{u}) \right) \\ &\quad - \frac{e_1}{4} \left(\sin(\Phi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,1}^1, \mathbf{u}) + \sin(\Xi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,2}^1, \mathbf{u}) \right. \\ &\quad \left. - \sin(\Pi_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^1, \mathbf{u}) - \sin(\Lambda_1) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^1, \mathbf{u}) \right); \end{aligned} \quad (26)$$

$$\begin{aligned}
G_{o,M_{s,2}^{w'}}^{A_1,A_2,A_3}(M_{s,2}^{w_0}f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(\cos(\Phi_2)G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,1}^2, \mathbf{u}) + \cos(\Xi_2)G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,2}^2, \mathbf{u}) \right. \\
&\quad \left. - \cos(\Pi_2)G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,3}^2, \mathbf{u}) - \cos(\Lambda_2)G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \\
&\quad + \frac{1}{4} \left(\sin(\Phi_2)G_{o,\phi}^{A_1,A_2,A_3'}f(\mathbf{w}_{s,1}^2, \mathbf{u}) + \sin(\Xi_2)G_{o,\phi}^{A_1,A_2,A_3'}f(\mathbf{w}_{s,2}^2, \mathbf{u}) \right. \\
&\quad \left. - \sin(\Pi_2)G_{o,\phi}^{A_1,A_2,A_3'}f(\mathbf{w}_{s,3}^2, \mathbf{u}) - \sin(\Lambda_2)G_{o,\phi}^{A_1,A_2,A_3'}f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \cdot e_2;
\end{aligned} \quad (27)$$

$$\begin{aligned}
G_{o,M_{s,3}^{w'}}^{A_1,A_2,A_3}(M_{s,3}^{w_0}f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,1}^3, u)e^{-e_4\Phi_3} + G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,2}^3, u)e^{-e_4\Xi_3} \right. \\
&\quad \left. - G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,3}^3, u)e^{-e_4\Pi_3} - G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,4}^3, u)e^{-e_4\Lambda_3} \right),
\end{aligned} \quad (28)$$

where $\Phi_i = \frac{d_i}{2b_i} \left[(2\pi b_i(w_0 - w'))^2 + 4\pi b_i w_i(w_0 - w') \right] + 2\pi u_i w'$; $\Xi_i = \frac{d_i}{2b_i} \left[(2\pi b_i(w_0 - w'))^2 - 4\pi b_i w_i(w_0 - w') \right] - 2\pi u_i w'$; $\Pi_i = \frac{d_i}{2b_i} \left[(2\pi b_i(w_0 + w'))^2 + 4\pi b_i w_i(w_0 + w') \right] - 2\pi u_i w'$; $\Lambda_i = \frac{d_i}{2b_i} \left[(2\pi b_i(w_0 + w'))^2 - 4\pi b_i w_i(w_0 + w') \right] + 2\pi u_i w'$; $A_i' = (a_i, -b_i, -c_i, d_i)$ ($i = 1, 2, 3$); $\mathbf{w}_{s,1}^1 = (w_1 + 2\pi b_1(w_0 - w'), w_2, w_3)$; $\mathbf{w}_{s,2}^1 = (w_1 - 2\pi b_1(w_0 - w'), w_2, w_3)$; $\mathbf{w}_{s,3}^1 = (w_1 + 2\pi b_1(w_0 + w'), w_2, w_3)$; $\mathbf{w}_{s,4}^1 = (w_1 - 2\pi b_1(w_0 + w'), w_2, w_3)$; $\mathbf{w}_{s,1}^2 = (w_1, w_2 + 2\pi b_2(w_0 - w'), w_3)$; $\mathbf{w}_{s,2}^2 = (w_1, w_2 - 2\pi b_2(w_0 - w'), w_3)$; $\mathbf{w}_{s,3}^2 = (w_1, w_2 + 2\pi b_2(w_0 + w'), w_3)$; $\mathbf{w}_{s,4}^2 = (w_1, w_2 - 2\pi b_2(w_0 + w'), w_3)$; $\mathbf{w}_{s,1}^3 = (w_1, w_2, w_3 + 2\pi b_3(w_0 - w'))$; $\mathbf{w}_{s,2}^3 = (w_1, w_2, w_3 - 2\pi b_3(w_0 - w'))$; $\mathbf{w}_{s,3}^3 = (w_1, w_2, w_3 + 2\pi b_3(w_0 + w'))$; $\mathbf{w}_{s,4}^3 = (w_1, w_2, w_3 - 2\pi b_3(w_0 + w'))$.

Proof. We will proof the formula (26). According to the definition of the OCSTLCT, we have

$$\begin{aligned}
G_{o,M_{s,1}^{w'}}^{A_1,A_2,A_3}(M_{s,1}^{w_0}f)(\mathbf{w}, \mathbf{u}) &= \int_{\mathbb{R}^3} M_{s,1}^{w_0}f(\mathbf{x}) \overline{M_{s,1}^{w'}\phi(\mathbf{x} - \mathbf{u})} \cdot K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x} \\
&= \int_{\mathbb{R}^3} f(\mathbf{x}) \sin(2\pi x_1 w_0) \overline{\phi(\mathbf{x} - \mathbf{u})} \sin(2\pi(x_1 - u_1)w') \cdot K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x} \\
&= -\frac{1}{4} \int_{\mathbb{R}^3} f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} (e^{-e_1 2\pi x_1 w_0} - e^{e_1 2\pi x_1 w_0}) \\
&\quad \cdot (e^{-e_1 2\pi(x_1 - u_1)w'} - e^{e_1 2\pi(x_1 - u_1)w'}) K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x} \\
&= -\frac{1}{4} \int_{\mathbb{R}^3} f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} (e^{-e_1 2\pi x_1(w_0 + w')} \cdot e^{e_1 2\pi u_1 w'} - e^{-e_1 2\pi x_1(w_0 - w')} \cdot e^{-e_1 2\pi u_1 w'} \\
&\quad - e^{-e_1 2\pi x_1(w' - w_0)} \cdot e^{e_1 2\pi u_1 w'} + e^{-e_1 2\pi x_1(-w' - w_0)} \cdot e^{-e_1 2\pi u_1 w'}) \cdot K_{A_1}^{e_1}(x_1, \omega_1) K_{A_2}^{e_2}(x_2, \omega_2) K_{A_3}^{e_4}(x_3, \omega_3) d\mathbf{x}.
\end{aligned}$$

According to Table 1, we obtain

$$((e^{-e_1 2\pi x_1 w_1} \cdot e_1) \cdot e^{-e_2 2\pi x_2 w_2}) \cdot e^{-e_4 2\pi x_3 w_3} = ((e^{-e_1 2\pi x_1 w_1} \cdot e^{e_2 2\pi x_2 w_2}) \cdot e^{e_4 2\pi x_3 w_3}) \cdot e_1.$$

Hence, we can obtain the formula (26).

Using the same method, (27) and (28) can be obtained. $\square$

Theorem 6. Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that for all $w_0, w' \in \mathbb{R}$, $\zeta_i = 2\pi w_0 x_i$, $\zeta'_i = 2\pi w' x_i$ ($i = 1, 2, 3$), $M_{c,1}^{w_0}f(x) = f(x) \cos \zeta_1$, $M_{c,1}^{w'}\phi(x) = \phi(x) \cos \zeta'_1$, $M_{c,2}^{w_0}f(x) = f(x) \cos \zeta_2$, $M_{c,2}^{w'}\phi(x) = \phi(x) \cos \zeta'_2$, $M_{c,3}^{w_0}f(x) = f(x) \cos \zeta_3$ and $M_{c,3}^{w'}\phi(x) = \phi(x) \cos \zeta'_3$. Then

$$\begin{aligned}
G_{o,M_{c,1}^{w'}}^{A_1,A_2,A_3}(M_{c,1}^{w_0}f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(e^{-e_1 \Phi_1} G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,1}^1, \mathbf{u}) + e^{-e_1 \Xi_1} G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,2}^1, \mathbf{u}) \right. \\
&\quad \left. + e^{-e_1 \Pi_1} G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,3}^1, \mathbf{u}) + e^{-e_1 \Lambda_1} G_{o,\phi}^{A_1,A_2,A_3}f(\mathbf{w}_{s,4}^1, \mathbf{u}) \right);
\end{aligned} \quad (29)$$

$$\begin{aligned}
G_{o, M_{c,2}^{w'}}^{A_1, A_2, A_3}(M_{c,2}^{w_0} f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(\cos(\Phi_2) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,1}^2, \mathbf{u}) + \cos(\Xi_2) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,2}^2, \mathbf{u}) \right. \\
&\quad \left. + \cos(\Pi_2) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) + \cos(\Lambda_2) G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \\
&\quad + \frac{1}{4} \left(\sin(\Phi_2) G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,1}^2, \mathbf{u}) + \sin(\Xi_2) G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,2}^2, \mathbf{u}) \right. \\
&\quad \left. + \sin(\Pi_2) G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) + \sin(\Lambda_2) G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \cdot e_2;
\end{aligned} \quad (30)$$

$$\begin{aligned}
G_{o, M_{c,3}^{w'}}^{A_1, A_2, A_3}(M_{c,3}^{w_0} f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{4} \left(G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,1}^3, u) e^{-e_4 \Phi_3} + G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,2}^3, u) e^{-e_4 \Xi_3} \right. \\
&\quad \left. + G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^3, u) e^{-e_4 \Pi_3} + G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^3, u) e^{-e_4 \Lambda_3} \right).
\end{aligned} \quad (31)$$

According to Theorem 5 and Theorem 6, we obtain the following theorem.

Theorem 7. Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that $w_0, w' \in \mathbb{R}$ and $M_{e,1}^{w_0} f(x) = f(x) e^{-e_1 2\pi w_0 x_1}$, $M_{e,1}^{w'} \phi(x) = \phi(x) e^{-e_1 2\pi w' x_1}$, $M_{e,2}^{w_0} f(x) = f(x) e^{-e_2 2\pi w_0 x_2}$, $M_{e,2}^{w'} \phi(x) = \phi(x) e^{-e_2 2\pi w' x_2}$, $M_{e,3}^{w_0} f(x) = f(x) e^{-e_4 2\pi w_0 x_3}$ and $M_{e,3}^{w'} \phi(x) = \phi(x) e^{-e_4 2\pi w' x_3}$. Then

$$G_{o, M_{e,1}^{w'}}^{A_1, A_2, A_3}(M_{e,1}^{w_0} f)(\mathbf{w}, \mathbf{u}) = e^{-e_1 \Pi_1} \cdot G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^1, \mathbf{u}); \quad (32)$$

$$\begin{aligned}
G_{o, M_{e,2}^{w'}}^{A_1, A_2, A_3}(M_{e,2}^{w_0} f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{2} \left[\cos(\Pi_2) \left(G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) - G_{o, \phi}^{A'_1, A_2, A_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) \right) \right. \\
&\quad \left. + \cos(\Lambda_2) \left(G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) + G_{o, \phi}^{A'_1, A_2, A_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \right] \\
&\quad + \frac{1}{2} \left[\sin(\Pi_2) \left(G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) - G_{o, \phi}^{A'_1, A_2, A'_3} f(\mathbf{w}_{s,3}^2, \mathbf{u}) \right) \right. \\
&\quad \left. + \sin(\Lambda_2) \left(G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) + G_{o, \phi}^{A'_1, A_2, A'_3} f(\mathbf{w}_{s,4}^2, \mathbf{u}) \right) \right] \cdot e_2;
\end{aligned} \quad (33)$$

$$\begin{aligned}
G_{o, M_{e,3}^{w'}}^{A_1, A_2, A_3}(M_{e,3}^{w_0} f)(\mathbf{w}, \mathbf{u}) &= \frac{1}{2} \left[\left(G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,4}^3, \mathbf{u}) - G_{o, \phi}^{A'_1, A'_2, A_3} f(\mathbf{w}_{s,4}^3, \mathbf{u}) \right) e^{-e_4 \Lambda_3} \right. \\
&\quad \left. + \left(G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_{s,3}^3, \mathbf{u}) + G_{o, \phi}^{A'_1, A'_2, A_3} f(\mathbf{w}_{s,3}^3, \mathbf{u}) \right) e^{-e_4 \Pi_3} \right].
\end{aligned} \quad (34)$$

Based on the Theorem 4 and Theorem 8, we can obtain the theorem of time-frequency shifts.

Theorem 8 (Time-frequency shifts). Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Operators of the form $M_{e,*}^* T_*$ is called time-frequency shifts. Then

$$G_{o, M_{e,1}^{w'} T_{r_1}'}^{A_1, A_2, A_3}(M_{e,1}^{w_0} T_{r_1} f)(\mathbf{w}, \mathbf{u}) = e^{-e_1 \Pi_1} \left(\cos \hat{\phi}_1^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\tilde{\mathbf{w}}_1, \mathbf{u}) - \sin \hat{\phi}_1^{A_1, A_2, A_3} \Theta_1^{A_1, A_2, A_3}(\tilde{\mathbf{w}}_1, \mathbf{u}) \right); \quad (35)$$

$$G_{o, M_{e,2}^{w'} T_{r_2}'}^{A_1, A_2, A_3}(M_{e,2}^{w_0} T_{r_2} f)(\mathbf{w}, \mathbf{u}) = \frac{1}{2} [(\cos \Pi_2 \cdot \odot + \cos \Lambda_2 \cdot \oplus) + (\sin \Pi_2 \cdot \otimes + \sin \Lambda_2 \cdot \vartheta) e_2], \quad (36)$$

$$G_{o, M_{e,3}^{w'} T_{r_3}'}^{A_1, A_2, A_3}(M_{e,3}^{w_0} T_{r_3} f)(\mathbf{w}, \mathbf{u}) = \frac{1}{2} (\triangle \cdot e^{-e_4 \Lambda_3} + \nabla \cdot e^{-e_4 \Pi_3}). \quad (37)$$

where $\hat{\phi}_i^{A_1, A_2, A_3} = r_i(w_i + 2\pi b_i(w_0 + w')c_i - \frac{a_i c_i r_i^2}{2})$, $\hat{\phi}'_i^{A_1, A_2, A_3} = r_i(w_i - 2\pi b_i(w_0 + w')c_i - \frac{a_i c_i r_i^2}{2})$, $\tilde{\mathbf{w}}_1 = (w_1 + 2\pi b_1(w_0 + w') - r_1 a_1, w_2, w_3)$, $\mathbf{w}_2 = (w_1, w_2 + 2\pi b_2(w_0 + w') - r_2 a_2, w_3)$, $\widehat{\mathbf{w}}'_2 = (w_1, w_2 - 2\pi b_2(w_0 + w') - r_2 a_2, w_3)$, $\mathbf{w}_3 = (w_1, w_2, w_3 - 2\pi b_3(w_0 + w') - r_3 a_3)$, $\widehat{\mathbf{w}}'_3 = (w_1, w_2, w_3 + 2\pi b_3(w_0 + w') - r_3 a_3)$,

$$\begin{aligned}\odot &= \cos \hat{\phi}_2^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_2, \mathbf{u}) - \sin \hat{\phi}_1^{A_1, A_2, A_3} \Theta_2^{A_1, A_2, A_3}(\mathbf{w}_2, \mathbf{u}) \\ &\quad - \cos \hat{\phi}_2^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\mathbf{w}_2, \mathbf{u}) + \sin \hat{\phi}_2^{A'_1, A'_2, A'_3} \Theta_2^{A'_1, A'_2, A'_3}(\mathbf{w}_2, \mathbf{u}); \\ \oplus &= \cos \hat{\phi}'_2^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\widehat{\mathbf{w}}'_2, \mathbf{u}) - \sin \hat{\phi}'_1^{A_1, A_2, A_3} \Theta_2^{A_1, A_2, A_3}(\widehat{\mathbf{w}}'_2, \mathbf{u}) \\ &\quad + \cos \hat{\phi}'_2^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\widehat{\mathbf{w}}'_2, \mathbf{u}) - \sin \hat{\phi}'_2^{A'_1, A'_2, A'_3} \Theta_2^{A'_1, A'_2, A'_3}(\widehat{\mathbf{w}}'_2, \mathbf{u}); \\ \otimes &= \cos \hat{\phi}_2^{A_1, A_2, A'_3} G_{o, \phi}^{A_1, A_2, A'_3} f(\mathbf{w}_2, \mathbf{u}) - \sin \hat{\phi}_1^{A_1, A_2, A'_3} \Theta_2^{A_1, A_2, A'_3}(\mathbf{w}_2, \mathbf{u}) \\ &\quad - \cos \hat{\phi}_2^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\mathbf{w}_2, \mathbf{u}) + \sin \hat{\phi}_2^{A'_1, A'_2, A'_3} \Theta_2^{A'_1, A'_2, A'_3}(\mathbf{w}_2, \mathbf{u}); \\ \uplus &= \cos \hat{\phi}'_2^{A_1, A_2, A'_3} G_{o, \phi}^{A_1, A_2, A'_3} f(\widehat{\mathbf{w}}'_2, \mathbf{u}) - \sin \hat{\phi}'_1^{A_1, A_2, A'_3} \Theta_2^{A_1, A_2, A'_3}(\widehat{\mathbf{w}}'_2, \mathbf{u}) \\ &\quad + \cos \hat{\phi}'_2^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\widehat{\mathbf{w}}'_2, \mathbf{u}) - \sin \hat{\phi}'_2^{A'_1, A'_2, A'_3} \Theta_2^{A'_1, A'_2, A'_3}(\widehat{\mathbf{w}}'_2, \mathbf{u}); \\ \triangle &= \cos \hat{\phi}_3^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}_3, \mathbf{u}) - \sin \hat{\phi}_3^{A_1, A_2, A_3} \Theta_3^{A_1, A_2, A_3}(\mathbf{w}_3, \mathbf{u}) \\ &\quad - \cos \hat{\phi}_3^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\mathbf{w}_3, \mathbf{u}) + \sin \hat{\phi}_3^{A'_1, A'_2, A'_3} \Theta_3^{A'_1, A'_2, A'_3}(\mathbf{w}_3, \mathbf{u}); \\ \nabla &= \cos \hat{\phi}'_3^{A_1, A_2, A_3} G_{o, \phi}^{A_1, A_2, A_3} f(\widehat{\mathbf{w}}'_3, \mathbf{u}) - \sin \hat{\phi}'_3^{A_1, A_2, A_3} \Theta_3^{A_1, A_2, A_3}(\widehat{\mathbf{w}}'_3, \mathbf{u}) \\ &\quad + \cos \hat{\phi}'_3^{A'_1, A'_2, A'_3} G_{o, \phi}^{A'_1, A'_2, A'_3} f(\widehat{\mathbf{w}}'_3, \mathbf{u}) - \sin \hat{\phi}'_3^{A'_1, A'_2, A'_3} \Theta_3^{A'_1, A'_2, A'_3}(\widehat{\mathbf{w}}'_3, \mathbf{u}).\end{aligned}$$

Theorem 9 [Inversion formula]. Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function, two real-valued time-domain window functions $\phi, \varphi \in L^2(\mathbb{R}^3) \setminus \{0\}$ and $\langle \phi, \varphi \rangle \neq 0$. Then we have the inversion formula of the OSTLCT

$$f(\mathbf{x}) = \frac{1}{\langle \phi, \varphi \rangle} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) K_{A_3^{-1}}^{e_4}(\omega_3, x_3) K_{A_2^{-1}}^{e_2}(\omega_2, x_2) K_{A_1^{-1}}^{e_1}(\omega_1, x_1) \varphi(\mathbf{x} - \mathbf{u}) d\mathbf{w} d\mathbf{u}. \quad (38)$$

Proof. Multiplying both sides of (9) from the right by $\varphi(\mathbf{x} - \mathbf{u})$ and integrating with respect to $d\mathbf{u}$ we get

$$\int_{\mathbb{R}^3} f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})} \varphi(\mathbf{x} - \mathbf{u}) d\mathbf{u} = \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) K_{A_3^{-1}}^{e_4}(\omega_3, x_3) K_{A_2^{-1}}^{e_2}(\omega_2, x_2) K_{A_1^{-1}}^{e_1}(\omega_1, x_1) \varphi(\mathbf{x} - \mathbf{u}) d\mathbf{w} d\mathbf{u} \quad (39)$$

Hence,

$$f(\mathbf{x}) \langle \phi, \varphi \rangle = \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) K_{A_3^{-1}}^{e_4}(\omega_3, x_3) K_{A_2^{-1}}^{e_2}(\omega_2, x_2) K_{A_1^{-1}}^{e_1}(\omega_1, x_1) \varphi(\mathbf{x} - \mathbf{u}) d\mathbf{w} d\mathbf{u}, \quad (40)$$

which completes the proof. $\square$

Theorem 10 [Plancherel theorem]. Let $f \in L^2(\mathbb{R}^3)$ be a real-valued functions and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Then we have

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u})|^2 d\mathbf{w} d\mathbf{u} = \frac{1}{2\pi|b_3|} \|\phi\|_{L^2(\mathbb{R}^3)}^2 \|f\|_{L^2(\mathbb{R}^3)}^2. \quad (41)$$

Proof. Let $h(\mathbf{x}, \mathbf{u}) = f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})}$. According to Lemma 2, we have

$$\int_{\mathbb{R}^3} |h(\mathbf{x}, \mathbf{u})|^2 d\mathbf{x} = 2\pi|b_3| \int_{\mathbb{R}^3} |\mathcal{L}_{A_1,A_2,A_3}^o \{h\}|^2 d\mathbf{w}. \quad (42)$$

Integrating both sides with respect to $d\mathbf{u}$ in (42), we obtain

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(\mathbf{x}) \overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{w} d\mathbf{u} = 2\pi|b_3| \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u})|^2 d\mathbf{w} d\mathbf{u}. \quad (43)$$

Hence,

$$\|\phi\|_{L^2(\mathbb{R}^3)}^2 \|f\|_{L^2(\mathbb{R}^3)}^2 = 2\pi|b_3| \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u})|^2 d\mathbf{w} d\mathbf{u}. \quad (44)$$

□

4 | HEISENBERG UNCERTAINTY PRINCIPLE FOR THE QWLCT

The Heisenberg uncertainty principle and quaternions are both basic for quantum mechanics. In quantum mechanics an uncertainty principle asserts that one cannot make certain of the position and velocity of an electron (or any particle) at the same time. That is, increasing the knowledge of the position decreases the knowledge of the velocity or momentum of an electron. In signal processing an uncertainty principle states that the product of the variances of the signal in the time and frequency domains has a lower bound. The Heisenberg uncertainty principles of the QSTLCT has been studied.²⁴ Recently, we³⁸ established the Heisenberg uncertainty principles associated with the OCLCT. Let us give a short and simple proof of the Heisenberg uncertainty principle of the OCLCT.

Lemma 3 [OCLCT uncertainty principle].³⁸ Let $f \in L^2(\mathbb{R}^3)$ and $\mathcal{L}_{A_1,A_2,A_3}^o \{f\}(\mathbf{w}) \in L^2(\mathbb{R}^3)$ then we have

$$\int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^3} \mathbf{w}^2 |\mathcal{L}_{A_1,A_2,A_3}^o \{f\}(\mathbf{w})|^2 d\mathbf{w} \geq \frac{2}{\pi|b_3|} b_1^2 b_1^2 \left(\int_{\mathbb{R}^3} |f(\mathbf{x})|^2 d\mathbf{x} \right)^2. \quad (45)$$

Now we arrive at the following important result.

Theorem 11 [OCSTLCT uncertainty principle]. Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a window function and $G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) \in L^2(\mathbb{R}^3)$ be the OCSTLCT of f . Then for every $f \in L^2(\mathbb{R}^3)$ we have the following inequality

$$\left(\int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \mathbf{w}^2 |G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u})|^2 d\mathbf{w} d\mathbf{u} \right)^{\frac{1}{2}} \geq \left(\frac{2}{\pi|b_3|} \right)^{\frac{1}{2}} |b_1 b_1| \|f\|_{L^2(\mathbb{R}^3)}^2 \|\phi\|_{L^2(\mathbb{R}^3)}. \quad (46)$$

Proof. Let $h(\mathbf{x}, \mathbf{u}) = f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}$. From Lemma 3, we have

$$\int_{\mathbb{R}^3} \mathbf{x}^2 |h(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^3} \mathbf{w}^2 \left| \mathcal{L}_{A_1, A_2, A_3}^o \{h\}(\mathbf{w}) \right|^2 d\mathbf{w} \geq \frac{2}{\pi |b_3|} b_1^2 b_1^2 \left(\int_{\mathbb{R}^3} |h(\mathbf{x})|^2 d\mathbf{x} \right)^2. \quad (47)$$

Taking the square root on both sides of (47), we obtain

$$\left(\int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^3} \mathbf{w}^2 \left| G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) \right|^2 d\mathbf{w} \right)^{\frac{1}{2}} \geq \left(\frac{2}{\pi |b_3|} \right)^{\frac{1}{2}} |b_1 b_1| \int_{\mathbb{R}^3} |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x}. \quad (48)$$

Integrating both sides with respect to $d\mathbf{u}$, we obtain

$$\int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^3} \mathbf{w}^2 \left| G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) \right|^2 d\mathbf{w} \right)^{\frac{1}{2}} d\mathbf{u} \geq \left(\frac{2}{\pi |b_3|} \right)^{\frac{1}{2}} |b_1 b_1| \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} d\mathbf{u}. \quad (49)$$

Furthermore, applying the Cauchy–Schwarz inequality to the left-hand side of (49), we have

$$\left(\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} d\mathbf{u} \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \mathbf{w}^2 \left| G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) \right|^2 d\mathbf{w} d\mathbf{u} \right)^{\frac{1}{2}} \geq \left(\frac{2}{\pi |b_3|} \right)^{\frac{1}{2}} |b_1 b_1| \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(\mathbf{x})\overline{\phi(\mathbf{x} - \mathbf{u})}|^2 d\mathbf{x} d\mathbf{u}. \quad (50)$$

Hence,

$$\|\phi\|_{L^2(\mathbb{R}^3)} \left(\int_{\mathbb{R}^3} \mathbf{x}^2 |f(\mathbf{x})|^2 d\mathbf{x} \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \mathbf{w}^2 \left| G_{o, \phi}^{A_1, A_2, A_3} f(\mathbf{w}, \mathbf{u}) \right|^2 d\mathbf{w} d\mathbf{u} \right)^{\frac{1}{2}} \geq \left(\frac{2}{\pi |b_3|} \right)^{\frac{1}{2}} |b_1 b_1| \|f\|_{L^2(\mathbb{R}^3)}^2 \|\phi\|_{L^2(\mathbb{R}^3)}^2. \quad (51)$$

We obtain the desired result. $\square$

5 | EXAMPLE OF THE OCSTLCT

Given the window function of the three-dimensional Haar function defined by

$$\phi(\mathbf{x}) = \begin{cases} 1, & 0 \leq x_1 < \frac{1}{2}, 0 \leq x_2 < \frac{1}{2}, 0 \leq x_3 < \frac{1}{2} \\ -1, & \frac{1}{2} \leq x_1 < 1, \frac{1}{2} \leq x_2 < 1, \frac{1}{2} \leq x_3 < 1 \\ 0, & \text{otherwise,} \end{cases} \quad (52)$$

Consider a 3D Gaussian quaternionic function of the form $f(x_1, x_2, x_3) = e^{-(\alpha_1 x_1^2 + \alpha_2 x_2^2 + \alpha_3 x_3^2)}$, for $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ are positive real constants.

From (7), then the OCSTLCT of f is given by

$$\begin{aligned}
 G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) &= B \int_{u_1}^{\frac{1}{2}+u_1} e^{-\alpha_1 x_1^2} e^{e_1 \left(\frac{a_1}{2b_1} x_1^2 - \frac{x_1 w_1}{b_1} + \frac{d_1}{2b_1} w_1^2 - \frac{\pi}{2} \right)} dx_1 \int_{u_2}^{\frac{1}{2}+u_2} e^{-\alpha_2 x_2^2} e^{e_2 \left(\frac{a_2}{2b_2} x_2^2 - \frac{x_2 w_2}{b_2} + \frac{d_2}{2b_2} w_2^2 - \frac{\pi}{2} \right)} dx_2 \\
 &\times \int_{u_3}^{\frac{1}{2}+u_3} e^{-\alpha_3 x_3^2} e^{e_4 \left(\frac{a_3}{2b_3} x_3^2 - \frac{x_3 w_3}{b_3} + \frac{d_2}{2b_3} w_3^2 - \frac{\pi}{2} \right)} dx_3 - B \int_{\frac{1}{2}+u_1}^{1+u_1} e^{-\alpha_1 x_1^2} e^{e_1 \left(\frac{a_1}{2b_1} x_1^2 - \frac{x_1 w_1}{b_1} + \frac{d_1}{2b_1} w_1^2 - \frac{\pi}{2} \right)} dx_1 \\
 &\times \int_{\frac{1}{2}+u_2}^{1+u_2} e^{-\alpha_2 x_2^2} e^{e_2 \left(\frac{a_2}{2b_2} x_2^2 - \frac{x_2 w_2}{b_2} + \frac{d_2}{2b_2} w_2^2 - \frac{\pi}{2} \right)} dx_2 \int_{\frac{1}{2}+u_3}^{1+u_3} e^{-\alpha_3 x_3^2} e^{e_4 \left(\frac{a_3}{2b_3} x_3^2 - \frac{x_3 w_3}{b_3} + \frac{d_2}{2b_3} w_3^2 - \frac{\pi}{2} \right)} dx_3
 \end{aligned} \quad (53)$$

By completing squares, we have

$$\begin{aligned}
 G_{\phi}^{A_1,A_2} f(\mathbf{w}, \mathbf{u}) &= B \int_{u_1}^{\frac{1}{2}+u_1} e^{-\left(\sqrt{\frac{2b_1 a_1 - e_1 a_1}{2b_1}} x_1 + \frac{e_1 \omega_1}{\sqrt{2b_1(2b_1 a_1 - e_1 a_1)}} \right)^2} e^{-\frac{\omega_1^2}{2b_1 a_1 - e_1 a_1} + \frac{e_1 d_1}{2b_1} \omega_1^2 - \frac{\pi}{2}} dx_1 \\
 &\times \int_{u_2}^{\frac{1}{2}+u_2} e^{-\left(\sqrt{\frac{2b_2 a_2 - e_2 a_2}{2b_2}} x_2 + \frac{e_2 \omega_2}{\sqrt{2b_2(2b_2 a_2 - e_2 a_2)}} \right)^2} e^{-\frac{\omega_2^2}{2b_2 a_2 - e_2 a_2} + \frac{e_2 d_2}{2b_2} \omega_2^2 - \frac{\pi}{2}} dx_2 \\
 &\times \int_{u_3}^{\frac{1}{2}+u_3} e^{-\left(\sqrt{\frac{2b_3 a_3 - e_4 a_3}{2b_3}} x_3 + \frac{e_4 \omega_3}{\sqrt{2b_3(2b_3 a_3 - e_4 a_3)}} \right)^2} e^{-\frac{\omega_3^2}{2b_3 a_3 - e_4 a_3} + \frac{e_4 d_3}{2b_3} \omega_3^2 - \frac{\pi}{2}} dx_3 \\
 &- B \int_{\frac{1}{2}+u_1}^{1+u_1} e^{-\left(\sqrt{\frac{2b_1 a_1 - e_1 a_1}{2b_1}} x_1 + \frac{e_1 \omega_1}{\sqrt{2b_1(2b_1 a_1 - e_1 a_1)}} \right)^2} e^{-\frac{\omega_1^2}{2b_1 a_1 - e_1 a_1} + \frac{e_1 d_1}{2b_1} \omega_1^2 - \frac{\pi}{2}} dx_1 \\
 &\times \int_{\frac{1}{2}+u_2}^{1+u_2} e^{-\left(\sqrt{\frac{2b_2 a_2 - e_2 a_2}{2b_2}} x_2 + \frac{e_2 \omega_2}{\sqrt{2b_2(2b_2 a_2 - e_2 a_2)}} \right)^2} e^{-\frac{\omega_2^2}{2b_2 a_2 - e_2 a_2} + \frac{e_2 d_2}{2b_2} \omega_2^2 - \frac{\pi}{2}} dx_2 \\
 &\times \int_{\frac{1}{2}+u_3}^{1+u_3} e^{-\left(\sqrt{\frac{2b_3 a_3 - e_4 a_3}{2b_3}} x_3 + \frac{e_4 \omega_3}{\sqrt{2b_3(2b_3 a_3 - e_4 a_3)}} \right)^2} e^{-\frac{\omega_3^2}{2b_3 a_3 - e_4 a_3} + \frac{e_4 d_3}{2b_3} \omega_3^2 - \frac{\pi}{2}} dx_3
 \end{aligned} \quad (54)$$

Making the substitutions $A_1 = \sqrt{\frac{2b_1 a_1 - e_1 a_1}{2b_1}}, A_2 = \sqrt{\frac{2b_2 a_2 - e_2 a_2}{2b_2}}, A_3 = \sqrt{\frac{2b_3 a_3 - e_4 a_3}{2b_3}}, B_1 = \frac{e_1 \omega_1}{\sqrt{2b_1(2b_1 a_1 - e_1 a_1)}}, B_2 = \frac{e_2 \omega_2}{\sqrt{2b_2(2b_2 a_2 - e_2 a_2)}}, B_3 = \frac{e_4 \omega_3}{\sqrt{2b_3(2b_3 a_3 - e_4 a_3)}}$, $J_1 = e^{-\frac{\omega_1^2}{2b_1 a_1 - e_1 a_1} + \frac{e_1 d_1}{2b_1} \omega_1^2 - \frac{\pi}{2}}, J_2 = e^{-\frac{\omega_2^2}{2b_2 a_2 - e_2 a_2} + \frac{e_2 d_2}{2b_2} \omega_2^2 - \frac{\pi}{2}}$ and $J_3 = e^{-\frac{\omega_3^2}{2b_3 a_3 - e_4 a_3} + \frac{e_4 d_3}{2b_3} \omega_3^2 - \frac{\pi}{2}}$ in the above expression we immediately obtain

$$\begin{aligned}
 G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) &= B \int_{u_1}^{\frac{1}{2}+u_1} e^{-(A_1 x_1 + B_1)^2} J_1 dx_1 \int_{u_2}^{\frac{1}{2}+u_2} e^{-(A_2 x_2 + B_2)^2} J_2 dx_2 \int_{u_3}^{\frac{1}{2}+u_3} e^{-(A_3 x_3 + B_3)^2} J_3 dx_3 \\
 &- B \int_{\frac{1}{2}+u_1}^{1+u_1} e^{-(A_1 x_1 + B_1)^2} J_1 dx_1 \int_{\frac{1}{2}+u_2}^{1+u_2} e^{-(A_2 x_2 + B_2)^2} J_2 dx_2 \int_{\frac{1}{2}+u_3}^{1+u_3} e^{-(A_3 x_3 + B_3)^2} J_3 dx_3
 \end{aligned} \quad (55)$$

Substituting $y_1 = A_1x_1 + B_1$, $y_2 = A_2x_2 + B_2$ and $y_3 = A_3x_3 + B_3$ in the above equation, we have

$$\begin{aligned}
 G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) &= \frac{J_1}{A_1\sqrt{2\pi b_1}} \int_{A_1u_1+B_1}^{A_1\left(\frac{1}{2}+u_1\right)+B_1} e^{-y_1^2} dy_1 \times \frac{J_2}{A_2\sqrt{2\pi b_2}} \int_{A_2u_2+B_2}^{A_2\left(\frac{1}{2}+u_2\right)+B_2} e^{-y_2^2} dy_2 \times \frac{J_3}{A_3\sqrt{2\pi b_3}} \int_{A_3u_3+B_3}^{A_3\left(\frac{1}{2}+u_3\right)+B_3} e^{-y_3^2} dy_3 \\
 &\quad - \frac{J_1}{A_1\sqrt{2\pi b_1}} \int_{A_1\left(\frac{1}{2}+u_1\right)+B_1}^{A_1(1+u_1)+B_1} e^{-y_1^2} dx_1 \times \frac{J_2}{A_2\sqrt{2\pi b_2}} \int_{A_2\left(\frac{1}{2}+u_2\right)+B_2}^{A_2(1+u_2)+B_2} e^{-y_2^2} dx_2 \times \frac{J_3}{A_3\sqrt{2\pi b_3}} \int_{A_3\left(\frac{1}{2}+u_3\right)+B_3}^{A_3(1+u_3)+B_3} e^{-y_3^2} dx_3 \\
 &= \frac{J_1}{A_1\sqrt{2\pi b_1}} \left(\int_0^{A_1u_1+B_1} (-e^{-y_1^2}) dy_1 + \int_0^{A_1\left(\frac{1}{2}+u_1\right)+B_1} e^{-y_1^2} dy_1 \right) \frac{J_2}{A_2\sqrt{2\pi b_2}} \\
 &\quad \times \left(\int_0^{A_2u_2+B_2} (-e^{-y_2^2}) dy_2 + \int_0^{A_2\left(\frac{1}{2}+u_2\right)+B_2} e^{-y_2^2} dy_2 \right) \\
 &\quad \times \frac{J_3}{A_3\sqrt{2\pi b_3}} \left(\int_0^{A_3u_3+B_3} (-e^{-y_3^2}) dy_3 + \int_0^{A_3\left(\frac{1}{2}+u_3\right)+B_3} e^{-y_3^2} dy_3 \right) \\
 &\quad - \frac{J_1}{A_1\sqrt{2\pi b_1}} \left(\int_0^{A_1\left(\frac{1}{2}+u_1\right)+B_1} (-e^{-y_1^2}) dy_1 + \int_0^{A_1(1+u_1)+B_1} e^{-y_1^2} dy_1 \right) \\
 &\quad \times \frac{J_2}{A_2\sqrt{2\pi b_2}} \times \left(\int_0^{A_2\left(\frac{1}{2}+u_2\right)+B_2} (-e^{-y_2^2}) dy_2 + \int_0^{A_2(1+u_2)+B_2} e^{-y_2^2} dy_2 \right) \frac{J_3}{A_3\sqrt{2\pi b_3}} \\
 &\quad \times \left(\int_0^{A_3\left(\frac{1}{2}+u_3\right)+B_3} (-e^{-y_3^2}) dy_3 + \int_0^{A_3(1+u_3)+B_3} e^{-y_3^2} dy_3 \right)
 \end{aligned} \tag{56}$$

Equation (56) can be written in the form

$$\begin{aligned}
 G_{o,\phi}^{A_1,A_2,A_3} f(\mathbf{w}, \mathbf{u}) &= \frac{J_1}{2\sqrt{2b_1\alpha_1 - e_1\alpha_1}} \left(-erf(A_1u_1 + B_1) + erf\left(A_1\left(\frac{1}{2} + u_1\right) + B_1\right) \right) \\
 &\quad \times \frac{J_2}{2\sqrt{2b_2\alpha_2 - e_2\alpha_2}} \left(-erf(A_2u_2 + B_2) + erf\left(A_2\left(\frac{1}{2} + u_2\right) + B_2\right) \right) \\
 &\quad \times \frac{J_3}{2\sqrt{2b_3\alpha_3 - e_4\alpha_3}} \left(-erf(A_3u_3 + B_3) + erf\left(A_3\left(\frac{1}{2} + u_3\right) + B_3\right) \right) \\
 &\quad - \frac{J_1}{2\sqrt{2b_1\alpha_1 - e_1\alpha_1}} \left(-erf\left(A_1\left(\frac{1}{2} + u_1\right) + B_1\right) + erf(A_1(1 + u_1) + B_1) \right) \\
 &\quad \times \frac{J_2}{2\sqrt{2b_2\alpha_2 - e_2\alpha_2}} \left(-erf\left(A_2\left(\frac{1}{2} + u_2\right) + B_2\right) + erf(A_2(1 + u_2) + B_2) \right) \\
 &\quad \times \frac{J_3}{2\sqrt{2b_3\alpha_3 - e_4\alpha_3}} \left(-erf\left(A_3\left(\frac{1}{2} + u_3\right) + B_3\right) + erf(A_3(1 + u_3) + B_3) \right)
 \end{aligned} \tag{57}$$

where $erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.

The STLCT plays a vital role in the analysis of signals. The windowed linear canonical kernel provides a unique representation for the signals so that the STLCT is highly effective. It is natural to think whether the OCSTLCT can also be applied to such problems. For illustrative purposes, we give the example above.

The potential applications can be found in the estimation of the lower bound of integral. Such as if a signal in Theorem 11 is determined, it is difficult to obtain the value of the left side of Theorem 11 by calculating the integral. However, an estimation of the left side of Theorem 11 can be easily obtained based on Theorem 11. Other uncertainty principles of the OCSTLCT can be studied,²⁶ such as the logarithmic uncertainty principle, the entropic uncertainty principle, the Lieb uncertainty principle and Donoho Stark's uncertainty principle. The Donoho Stark's uncertainty principle can be applied in signal recovery.¹³ Moreover, based on the OCSTLCT, it can provide convenience for studying the uncertainty principles of the other form transforms, such as Wigner distribution (WD) and ambiguity function (AF).³⁹ Other potential applications can be found, such as the 3D edge detection of gray images,²¹ swept-frequency filters,²³ and detection of octonion-valued linear frequency-modulated signal.³⁹

6 | CONCLUSIONS

In this paper, using the basic concepts of octonion algebra, we proposed the OCSTLCT. Important properties of the OCSTLCT such as boundedness, linearity, parity, shift, modulation, inversion formula, and orthogonality relation were derived. In addition, based on the properties associated with OCSTLCT and the uncertainty principle for the OCLCT, a new uncertainty principle for the OCSTLCT was established. Finally, an example was given. The results in this paper are new in the literature.

ACKNOWLEDGEMENT

This work was funded by grants from National Key Research and Development Project of China (Grant No. 2020YFC2006205).

CONFLICT OF INTEREST

This work does not have any conflicts of interest.

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How to cite this article: Gao W-B, Li B-Z. Octonion short-time linear canonical transform. *Math Meth Appl Sci.* 2023;46(1):321-338. doi:10.1002/mma.8513